MIL-M-63044A (MI)
15 MAY 1980

Superseding
MIL-M-63044 (TM)
2 September 1970

MILITARY SPECIFICATION
MANUALS, TECHNICAL: MISSILE SYSTEM EQUIPMENT UNIT-UNDER-TEST (UUT) PROCEDURES

This specification is approved for use by Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1. **Scope.** This specification contains the detail requirements for preparing technical manuals on unit-under-test (UUT) procedures (6.3*a*) for missile system equipment.

1.2. **Classification.** The type of technical manual to be prepared in accordance with this specification is type -30, Direct Support, -34, Direct Support and General Support, or -40, General Support UUT Maintenance Manuals.

2. APPLICABLE DOCUMENTS

2.1. The following documents of the issue in effect on the date of invitation for bid, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-38784	—Manuals, Technical: General Requirements for Preparation of
MIL-M-63038	—Manuals, Technical: General Requirements for Manuscripts

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1. **General.**

3.1.1. *Content.* Content shall be as specified herein except that revisions to existing manuals shall be prepared as specified by the procuring activity.

3.2. *UUT maintenance manual.* UUT maintenance manuals include procedures for performing maintenance allocated to DS and/or GS maintenance by the maintenance allocation chart. The arrangement of the manual shall be as follows:

Front Matter
Chapter 1 - Introduction
Chapter 2 — (Insert nomenclature and identifying number of UUT)

3.2.1. *Front matter.* See MIL-M-38784 and MIL-M-63038.

3.2.2. *Chapter 1.* Prepare appropriate chapter 1 as required by MIL-M-63038.

3.2.2.1. *Scope.* This paragraph shall explain the arrangement of the manual and the format of the UUT chapters. Reference shall be made to the related end item manual(s).

3.2.2.2. *Maintenance forms and records.* This paragraph shall include the following statement: "Department of the Army forms and procedures used for equipment maintenance will be those prescribed by TM 38-750."

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to:
Commander, US Army Missile Command, ATTN:DRSMI-NP, Redstone Arsenal, AL 35809, by using the self-addressed Standardization Document Improvement Proposal(DD Form 1426) appearing at the end of this document or by ltr.

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3.2.3. *Chapter 2 — (Insert nomenclature and identifying number of UUT procedures.)*

3.2.3.1. *General.*

3.2.3.1.1. *Coverage.* A separate chapter shall be prepared for each significantly different UUT (6.3*a*) to be tested at the shop set.

3.2.3.1.2. *Differences between models.* Minor differences shall be explained within the body of the UUT procedures and summarized in a paragraph preceding the UUT procedures within the chapter. Separate chapters shall be prepared for each configuration when differences are significant.

3.2.3.2. *Format.* The format for UUT procedures shall be similar to the samples in figure 1.

3.2.3.3. *Content.*

3.2.3.3.1. *Preparation for test.* This paragraph shall contain all instructions necessary to prepare the UUT for the dynamic (hot) test. Switch positions shall be identified for all controls on the UUT and on special test equipment. An illustration of the UUT showing location of the required controls and test points shall be included in this paragraph. A table shall be included to identify all cables, test equipment, and test accessories required to perform the UUT procedures. When required, test set-up diagrams shall be prepared.

3.2.3.3.2. *Description of test.* This paragraph shall be limited to that data absolutely required to perform the UUT procedures. When a description of the test is required, it shall be limited to explanations of inputs, outputs, test points, controls, indicators, and circuit descriptions sufficient to understand troubleshooting to the level of those repair parts prescribed for replacement by DS or GS personnel. Circuit description shall not be included for parts not replaced by DS/GS or lower level of maintenance. Specific values of inputs, outputs, and test points shall be given in the UUT procedure and schematic diagrams as applicable.

3.2.3.3.3. *Schematic diagrams.* Schematics shall be used where necessary to support the theory of test and the UUT procedures. Parts such as printed circuit boards, modules, chips, and encapsulated circuits which are replaced as a repair part shall be shown as a box. (See figure 2.) The part shall be identified by its functional name and reference designator. Brief representations of multi-stage or multi-function components may be provided within the box. For conventional hardwired UUT's, where parts are replaced, a complete electrical schematic shall be provided if required to perform the UUT procedures. (See figure 3.) All built-in test points, controls, indicators, and inputs and outputs identified in the UUT procedure shall be functionally represented.

3.2.3.3.4. *UUT procedures.* The UUT procedures shall be so constructed that any defective authorized repair part will be detected. Each complete functional test of a UUT shall be prepared as a separate table. Each subfunctional test within the table shall be numbered. For example, in figure 1, the balance test is step 1, the detector-gain test is step 2, and the summing gain test is step 3. Each step shall be in three parts: (1) position of all controls and procedures for obtaining a normal indication, (2) the normal indication, and (3) corrective action for abnormal indications. Reference designators for parts to be replaced shall be listed. The procedures shall be so arranged that they may be performed in minimum time. Avoid checking the same circuit or part more than once. Limit the checks to those absolutely necessary to identify the defective replaceable part.

3.2.3.3.4.1. *Complex corrective procedures.* When the design of a UUT is not complex, normal troubleshooting logic may be used as shown in figure 1. When the design of a UUT is complex, the troubleshooting information required to isolate to a defective component will usually be complex. The format for complex troubleshooting is shown in figure 4. Its logic is to isolate to the defective component, then set up the equipment to repeat the test where the failure was detected. Figure 4 has explanatory notes, shown in boxes, to explain the content and logic of the various parts of the format.

3.2.3.3.5. *Repair instructions.* This paragraph shall include all necessary instructions for repairing each defect identified by the UUT procedures. Line drawings shall be used as necessary to illustrate assembly and disassembly instructions. Maximum

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use shall be made of the artwork prepared for the list of parts illustrated (LPI) manuals. Coverage shall not be included for procedures which are obvious to the equipment technician for whom this manual is intended.

3.2.3.3.6. *Maintenance calibration instructions.* Normally these instructions are included as an integral part of the UUT procedures. This paragraph shall include any necessary supplemental instructions required to insure that the UUT has been maintenance calibrated prior to being returned to organizational or supply channels.

3.2.3.3.7. *Final acceptance tests.* This paragraph shall include any supplemental instructions necessary to insure that the UUT is serviceable and has passed all the standards identified in the UUT procedure. Normally, these instructions will repeat specific steps of the UUT and maintenance calibration procedures followed by specific supplemental procedures.

3.3. Validation and verification. (See MIL-M-63038.)

4. QUALITY ASSURANCE PROVISIONS

This section is not applicable to this specification.

5. PREPARATION FOR DELIVERY

(See MIL-M-38784.)

6. NOTES

6.1. Intended Use. The manuals prepared in accordance with this specification provide a source of instructions for the test and repair of missile system chassis and other assemblies at the maintenance levels indicated by the specific manual concerned.

6.2. Ordering Data. The procurement documents shall specify:

a. Title, number, and date of this specification.

b. Number and title of technical manual to be prepared.

6.3. Definitions.

a. UUT — Unit-under-test. An item of equipment, usually at the chassis level, that is removed from the end item and returned to a DS or GS shop set for troubleshooting and repair, commonly called a secondary item. Non-UUT is an item that is repaired while in the end item and is not returned to a DS or GS shop set for maintenance.

b. Console. The test equipment located at a test position where the UUT procedures are to be performed. The assemblage of test equipment can be rack mounted or portable or a combination of both. Consoles are usually numbered or named, i.e., console 1, console 2, RF console, CW console, high frequency console, etc.

c. Maintenance calibration. The process of comparing the accuracy of system test, measurement, or diagnostic equipment against other equipment that has been calibrated by a Department of the Army calibration team. Maintenance calibration is performed by the personnel who perform the UUT procedures.

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CHAPTER 0

DOPPLER AMPLIFIER – COMPARATOR-9196586

00. Preparation for Test.

a. Inspect uut for obvious defects.

b. Panel 2 and panel 3 circuit breakers - - - - - OFF.

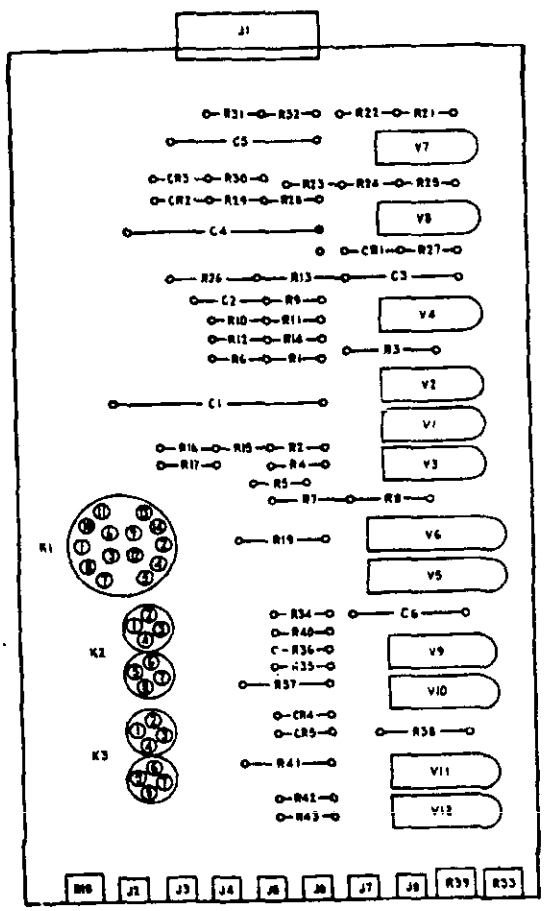


Figure 00. Doppler amplifier-comparator.

Table 0-00. Test Equipment

Cw/pulse console
Multimeter test lead stt (2)
Coaxial termination box CWR-570A
Test Lead
W-024 (3)
W-568A
10064044 (2)
10064045

c. TEST SELECTOR . . . . . OFF.

d. TEST . . . . . OFF.

e. PROGRAM . . . . . MAN.

f. Connect multimeter test lead sets to MULTIMETER A and B.

g. Multimeter A ac probe in programmer J2.

h. Multimeter B ac probe in programmer J3.

i. Insert CWR-570A.

j. MULTIMETER A and B . . . . . OVERRIDE.

k. Panel 2 and panel 3 circuit breakers . . . . . .ON.

l. TEST . . . . . .OFF.

m. Wait until console stabilizes.

n. Check pb jack voltages.

o. TEST . . . . . .OFF.

p. Connect W-568A between uut J1 and panel 5 J1.

q. PROGRAM . . . . . .AUTO.

r. TEST . . . . . ON.

Figure 1. Sample UUT format (sheet 1 of 3).

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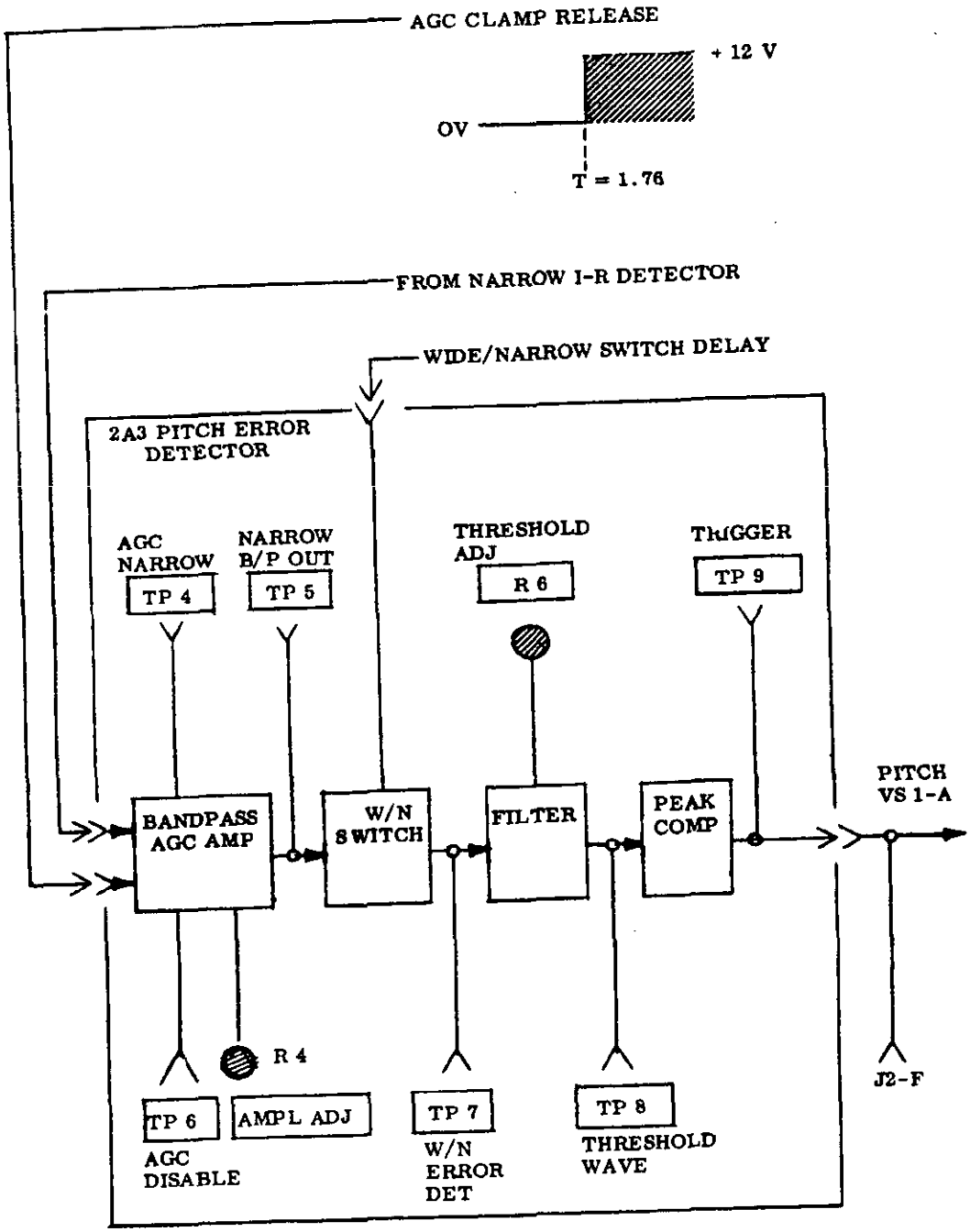


FIGURE 2. *Sample UUT schematic diagram.*

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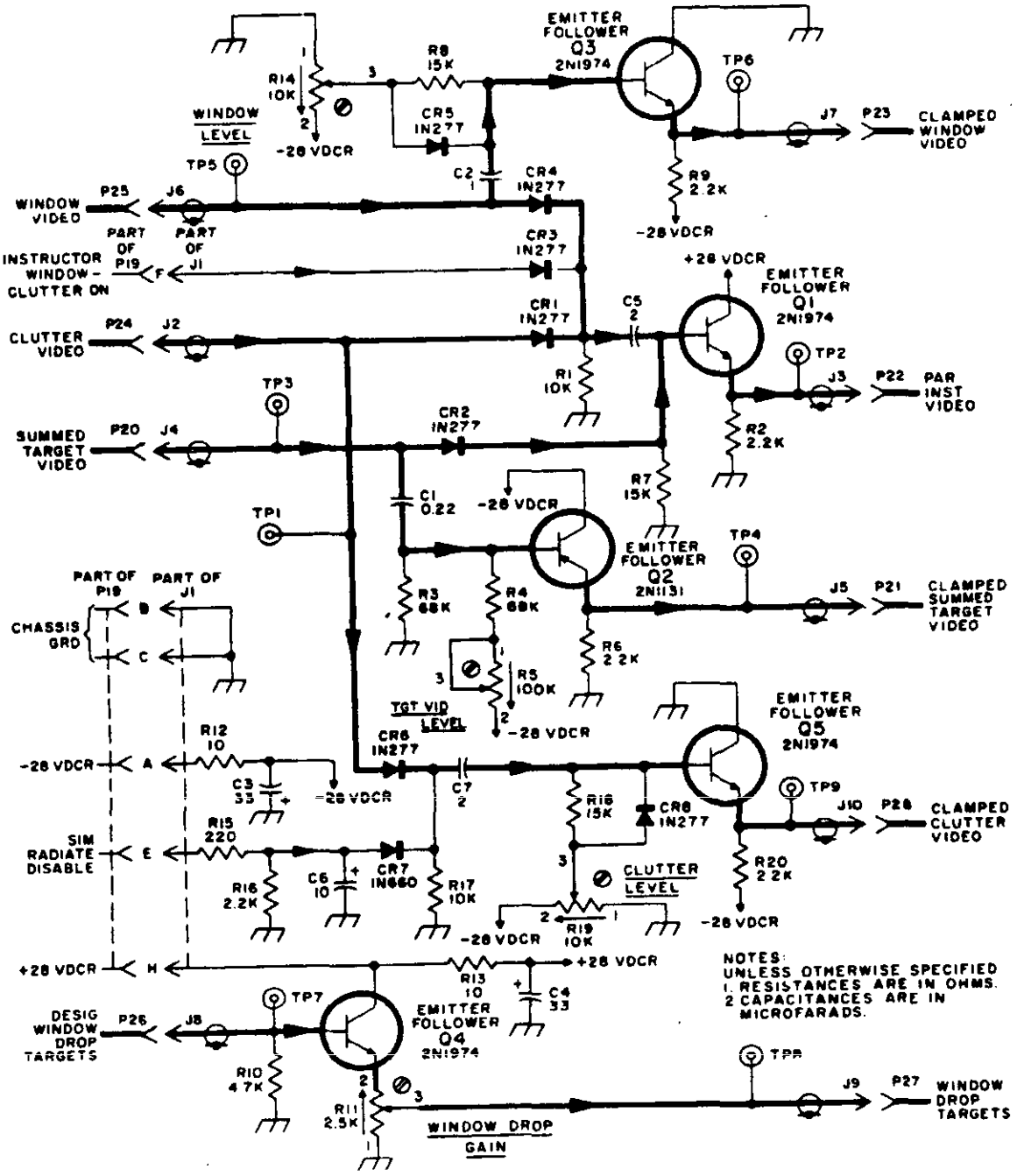


FIGURE 8. *Sample complete UUT schematic diagram.*

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Table 0-0. Frequency Counter — Periodic Test Procedures

Step	Operation	Normal indication	Corrective procedure
1.	AC Power		
a.	Multimeter A VAC ..... press.		
	Connect cable 1 to point 1.		Defective components shall not be listed if troubleshooting procedures are provided. The troubleshooting procedures should isolate the defective component.
	A1, A2		Set console/uut controls/switches. Use abbreviated uut format: example: A VOLTS/CM ..... 0.5. Multimeter A VDC ..... press. Insert multimeter A VDC probe into uut TP6.
	(1) (Action 1)		
	(Action 2)		Connect applicable test lead/cable/probe to console/unit.
		Indication	
	Return console test equipment to original configuration in order to repeat the test. Be specific. Use abbreviated uut format as indicated above.	If normal: A1, perform step 1a(2) below If abnormal: A2, perform step 1a(2) below	
	(2) After repair, repeat step 1a above.		

Figure 4. Complex corrective procedures format (sheet 1 of 2).

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Sample Table — Continued

Step	Operation / Normal indication / Corrective procedure
1b.	Multimeter A VAC . . . . . press. Place multimeter A AC VOLTS probe on uut 7A7T1-3. Indication 2 em → (1) (Action 3). Aline under "C" in Corrective Procedure → (Action r). 2 em indent → Indication If normal: A3, perform step 1b(4) below. 4 em indent → If abnormal: perform step 1b(2) below. (2) (Action 5). Last indent. Do NOT go below this level. Indication Indication If normal: A4, perform step 1b(4) below. If abnormal: perform step 1b(3) below.

Figure 4. Complex corrective procedures format (sheet 2 of 2).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-63044A(MI), Manuals, Technical: Missile System Equipment Unit-Under-Test (UUT) Procedures

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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